THE SET OF POINTS WHERE THE FIBRES ARE UNRAMIFIED IS COMPATIBLE WITH ARBITRARY BASE CHANGE

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 7937-7955 and proof span 7957-7977 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement displays the point set twice, once in terms of the fibres and once in terms of the morphism; both descriptions are the author's and are preserved.

Editorial note. The proof uses the author's notation for the base change and its projection; it is preserved unchanged.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The description of the fibre of a base change as the base change of the fibres, and the base change statement for the sheaf of relative differentials, cited from the same chapter. Cited at proof lines 7959-7969. Identifies the fibre differentials of the base change with the pullback of the fibre differentials.
- The pointwise characterization of being unramified in terms of the vanishing of the sheaf of relative differentials at the point. Cited at proof lines 7970-7971. Turns the comparison of differentials into the equality of the two point sets.
- Under the locally finitely presented hypothesis the same point set is the open set where the morphism is G-unramified, cited from the same chapter. Cited at proof lines 7972-7977. Extends the statement to the G-unramified version.

Source reference. Original lemma 01V0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2429–2429. Same-chapter reference used in the retained author text at source line(s) 6683, 7332, 7905, 7971. It is supplied only as

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a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02GF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3226–3226. Same-chapter reference used in the retained author text at source line(s) 7973, 7976, 8398, 8413. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume f locally of finite type. Formation of the open set*

$$\begin{aligned} T &= \{x \in X \mid X_{f(x)} \text{ is unramified over } \kappa(f(x)) \text{ at } x\} \\ &= \{x \in X \mid X \text{ is unramified over } S \text{ at } x\} \end{aligned}$$

commutes with arbitrary base change: For any morphism $g : S' \to S$, consider the base change $f' : X' \to S'$ of f and the projection $g' : X' \to X$. Then the corresponding set T' for the morphism f' is equal to $T' = (g')^{-1}(T)$. If f is assumed locally of finite presentation then the same holds for the open set of points where f is G-unramified.

Proof. Let $s' \in S'$ be a point, and let $s = g(s')$. Then we have

$$X'_{s'} = \mathrm{Spec}(\kappa(s')) \times_{\mathrm{Spec}(\kappa(s))} X_s$$

In other words the fibres of the base change are the base changes of the fibres. In particular

$$\Omega_{X_s/s,x} \otimes_{\mathcal{O}_{X_s,x}} \kappa(x') = \Omega_{X'_{s'}/s',x'} \otimes_{\mathcal{O}_{X'_{s'},x'}} \kappa(x')$$

see Lemma 01V0. Whence $x' \in T'$ if and only if $x \in T$ by Lemma 02GF. The second part follows from the first because in that case T is the (open) set of points where f is G-unramified according to Lemma 02GF. □